

Real-time adaptive perception

The latest lidar modules now feature an adaptive perception, which is the ability of lidar systems to adjust their perception parameters based on changing environmental conditions in real time. The performance of lidar systems can be affected by factors such as weather conditions, lighting, and the reflectivity of the surrounding objects.

Adaptive perception algorithms in lidar systems can dynamically adjust the lidar's parameters, such as laser power, range, and scan rate, based on the changing environment to optimise the performance of the system. For example, if the lidar is operating in foggy conditions, the system can reduce the laser power and increase the scan rate to improve the accuracy of the measurements.

Adaptive perception is particularly important for lidar systems used in autonomous vehicles, where the system must operate in a wide range of environmental conditions while maintaining accurate and reliable perception. By adjusting its parameters in real time, an adaptive perception lidar system can improve the safety and efficiency of autonomous vehicles by providing reliable and accurate environmental information. Velodyne Alpha Prime, Ouster OS Dome, and Rev 7 lidar are some of the lidars with real-time adaptive perception.

Better range

The lidar technology has been improving and there's a common trend of lidar having a higher range. Lidars such as RIEGL VZ-4000 can detect objects up to a range of 4,000 metres. Lidar modules for autonomous vehicles had a much lower range, but newer lidars like Aeries II offer a range of up to 500 metres. Vista-X120 or lidars featuring the Ouster REV7 sensors can offer a range of over 200 metres for autonomous vehicle applications.



Graphic representation of an autonomous vehicle's adaptive perception

To improve the detection range of lidars, manufacturers have increased their power and sensitivity. A higher range in lidar provides several advantages for various applications. It helps detect objects that are farther away from the sensor and thus reduce the risk of collisions. Moreover, the higher range can cover a larger area in a single scan, which can reduce the time required to scan an entire environment and provide more comprehensive coverage. A higher range also enables improved mapping accuracy. Furthermore, the higher range is useful for applications such as autonomous vehicles, robotics, and industrial automation.

More lidar sensors in a single module

Newer lidar modules feature multiple sensors. More sensors in each module offer advantages like improved coverage, redundancy, better accuracy, and a lower cost. Velodyne HDL-32E features 32 laser sensors while Ouster OS-1 packs 64 laser sensors in it. Having multiple sensors allows for a more comprehensive view of the environment. By placing sensors at different angles or heights, the system can obtain a

more complete 360-degree view of the surrounding area.

Higher number of sensors is particularly important in safety-critical applications, such as autonomous vehicles, where any failure of the sensor system could have serious consequences. Using multiple sensors can also improve the accuracy of the system. By combining the data from multiple sensors, the system can filter out noise and errors in individual sensor readings, resulting in a more accurate overall measurement.

Another advantage of having multiple sensors is that it can allow use of lower-cost sensors that may not provide the same level of performance individually as more expensive sensors. But by combining multiple lower-cost sensors, the system can achieve similar or better performance at a lower overall cost.

Solid-state lidars

Lidar technology is used in many harsh and difficult applications, including UAVs, vehicles, and even underwater search and mapping applications. To enhance the longevity of the devices, the newer lidars have fewer moving parts.

Solid-state lidars do not having